

Reasons for and against classifying neutrinos as Cosmic Rays

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Cosmic rays were historically identified through their strong ionization of the atmosphere, a criterion that still shapes their modern definition and excludes weakly interacting cosmic messengers such as neutrinos. I hypothesise that a minimum effective interaction requirement, rather than electric charge, provides a clearer and historically consistent boundary between primary cosmic rays and cosmic neutrinos. In this article, I briefly review the historical context of cosmic rays and compare the energy spectra, cross section, detection methods, origin, anisotropy, and electric charge. I conclude that, although they share similarities, their physical differences justify maintaining a distinct classification to avoid confusion.

I. INTRODUCTION

Cosmic rays (CRs) are not “rays” in the sense of a particle flux, but individual particles travelling so fast that their trajectories appear ray-like. They consist predominantly of protons ($\sim 89\%$), helium nuclei ($\sim 10\%$) and heavier nuclei ($\sim 1\%$), although some primaries consist of positrons and antiprotons. In the historical definition, which has prevailed to the present day, they are identified as high-energy charged particles of extraterrestrial origin. This definition emerged from early atmospheric ionization measurements, which implicitly tied the concept of CRs to charged particles that interact efficiently with matter primarily via the strong force. This charge- and interaction-based origin of the definition has persisted, even as new cosmic messengers, such as neutral neutrinos, have been observed.

This raises a conceptual problem: how to categorise energetic particles of cosmic origin. As the particles differ drastically in their interaction with matter, I propose that a minimum effective interaction criterion provides a more coherent basis for defining primary CRs than electric charge alone.

II. IONIZATION EVIDENCE FOR COSMIC RADIATION

From the late 18th to the late 19th century, scientists including C.-A. de Coulomb and later C. T. R Wilson, using electroscopes, revealed that air is not a perfect insulator. Charged electroscopes leaves slowly discharged due to the continuous creation of ions in air, quantified by the ionization rate, i.e., the number of ions created per unit volume and time.

Initially, this radiation was attributed to radioactive elements in the Earth’s crust. J. Elster and H. Geitel in 1899 [1] showed that enclosing an electroscope in a metal container did not prevent discharge, while evacuating

the air stopped it entirely. This demonstrated that ionization was caused by radiation acting on the air itself. Subsequent measurements by A. S. Eve [2] found a nearly constant ionization rate of about $10 \text{ ions cm}^{-3} \text{ s}^{-1}$, even far from radioactive sources.

Domenico Pacini [3] performed measurements over the sea and underwater, where radiation from the crust should be negligible. He first calculated the ionization expected from known radioactive sources in air, water, and the detector itself, finding a total contribution of only $\simeq 0.08 \text{ ions cm}^{-3} \text{ s}^{-1}$ to the ionization of enclosed air inside the electroscope.

However, measurements with a sealed electroscope, showed that the net contribution from external radiation was $\simeq 3.4 \text{ ions cm}^{-3} \text{ s}^{-1}$, dozens of times higher than what the theory predicted.

Pacini then compared measurements at the surface and at a depth of 3 m in water, finding a reduction of about $2.1\text{--}2.2 \text{ ions cm}^{-3} \text{ s}^{-1}$, consistent with the exponential attenuation law

$$I(d) = I_0 e^{-\lambda' d}, \quad (1)$$

where $I(d)$ is the radiation intensity after traversing a thickness d , I_0 the initial intensity, and λ' the absorption coefficient of the medium ($\lambda' \simeq 0.034 \text{ cm}^{-1}$ for water). These results concluded that a sizable cause of ionization exists in the atmosphere, independent of the direct action of radioactive substances in the soil.

Another confirmation came from balloon experiments by Gockel and Victor Hess. In 1912, Hess carried sealed electroscopes to altitudes of up to 5 km [4] and observed that ionization increased, reaching nearly twice the ground-level value.

Later flights by Kohlhörster [5] extended this trend to altitudes of 9 km. Measurements performed at night and during a solar eclipse ruled out the Sun as the primary source. The negative exponential trend ($I(d) < I_0$) proved that the flux of penetrating radiation enters the water from above, implying a cosmic origin.

Since then, it was established that a substantial component of atmospheric ionization originates from an extraterrestrial source, now known as cosmic rays.

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III. COSMIC RAYS AND NEUTRINOS

Throughout this work, the term cosmic rays refers to primary cosmic rays, that is, energetic particles of extraterrestrial origin before their first interaction with the Earth's atmosphere. Particles produced by the interaction of a primary CR with atmospheric nuclei are secondary cosmic rays, generated through air showers that develop via hadronic or electromagnetic cascades, and, later, also leptonic.

The atmospheric neutrino flux arises mainly from the two-body decays of pions and kaons and the subsequent muon decays. Therefore, neutrinos are part of secondary CRs; however, primary CRs, and thus the definition of cosmic rays, do not include them.

Neutrinos are classified as cosmic when they originate from astrophysical sources. As is well known, they are neutral and interact through the weak force. These were first detected in 1968 by Raymond Davis Jr. and his collaborators [6], who observed solar neutrinos through rare interaction events.

The definition of CRs takes into account their later interaction with the atmosphere, excluding neutrinos because their cross-section, the interaction probability, is too small. However, if considering no minimum interaction, energetic neutrinos could be classified as CRs and would produce cascades as well. Thus, when defining cosmic-ray interactions, a minimum effective cross-section must be assumed.

Both CRs and neutrinos are merged within the broader category of cosmic radiation, which also includes electromagnetic radiation. However, cosmic neutrinos are more closely related to CRs than to electromagnetic messengers. Clear lines between CRs and cosmic neutrinos help avoid confusion in data interpretation.

IV. COSMIC-RAY AND NEUTRINO COMPARISON

A. Energy spectra

When describing CRs, it is common to use their energy spectra. Their spectra are the distribution of CRs or neutrinos at each energy. Both are considered ultra-high-energy (UHE) above 10^{17} eV, although only one UHE neutrino has been detected so far. However, more neutrinos can theoretically reach UHE energies.

Most CRs are relativistic, with kinetic energies comparable to or exceeding their rest-mass energies. Their energy spectrum extends from 10^9 eV up to 10^{20} eV, therefore including some ultra-high-energy cosmic rays (UHECRs). However, detected cosmic neutrinos so far span energies from 10^{13} eV to 2×10^{15} eV,

as first reported by IceCube in 2013 [7], with a single UHE event at 2.2×10^{17} eV, observed by KM3NeT in 2025 [8], interpreted as the first possible cosmogenic neutrino detected. The next most energetic neutrinos are about 100 times less energetic.

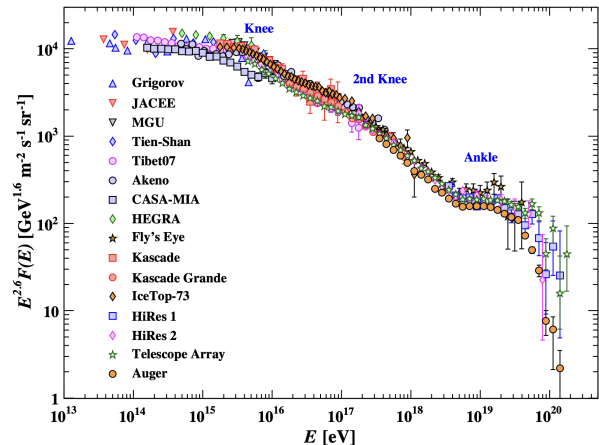


FIG. 1. The all-particle spectrum as a function of E (energy-per-nucleus) from air shower measurements [9].

Cosmic-ray spectra for $E \gtrsim 3 \times 10^{10}$ eV is typically a broken power-law, with characteristic features where the slope changes rapidly, as shown in Fig. 1. The first major feature is the knee at $\sim 3 \times 10^{15}$ eV, where the spectrum steepens ($\alpha \approx 2.7$ to 3.1). At the second knee, at $\sim 10^{17}$ eV, there is a milder further steepening (to $\alpha \approx 3.3$). Another feature is the ankle at $\sim 5 \times 10^{18}$ eV, where the spectrum flattens (the index decreases to $\alpha \approx 2.3$). Above $\sim 5 \times 10^{19}$ eV, the flux exhibits an exponential suppression identified as the Greisen–Zatsepin–Kuzmin (GZK) cutoff, caused by interactions of UHECRs with the Cosmic Microwave Background (CMB) and Extragalactic Background Light (EBL). Studying how fast the slope changes tells us what physical processes shape the cosmic-ray spectrum.

Neutrinos produced in cosmic-ray interactions get about 5% of the parent proton's energy. Because of this, the neutrino spectrum looks similar in shape to the cosmic-ray spectrum at the source, but shifted to lower energies. At still higher energies, there are theorized cosmogenic neutrinos expected from interactions of UHECRs with the CMB and the EBL, which are expected to appear at EeV energies.

The theorised flux of these is close to the Waxman–Bahcall level, which corresponds to a typical power-law spectrum produced by Fermi acceleration, $dN/dE \propto E^{-2}$ [10]. No strong spectral features have been observed yet, likely due to the limited data available, although a power-law slope could be expected since these neutrinos originate from cosmic-ray acceleration processes.

B. Cross section

In the case of hadronic interactions in CRs, which proceed via the strong force, the cross sections depend only weakly on energy, increasing slowly at higher energies. The proton–air cross section σ is large (of order ~ 500 mb [11]). The cross sections in the interstellar medium (inelastic p – p scattering, p –He scattering, and nuclear fragmentation processes of heavier nuclei) are also large, typically tens to hundreds of millibarns, as obtained from semi-empirical nuclear cross-section models fitted to accelerator data, such as those developed by Tripathi et al. and by Webber, Silberberg, and Tsao [12]. UHECRs interact with photon backgrounds, via processes characterized by their energy-loss lengths, the typical distance over which they lose a significant fraction of their energy: pair production has a loss length of several hundred megaparsecs (Mpc), while photopion production above $\sim 5 \times 10^{19}$ eV falls to tens of Mpc. For nuclei, photodisintegration dominates, with interaction lengths that can shrink to only a few Mpc [13].

Regarding cosmic-ray electrons, propagation in the Galaxy is dominated by energy losses rather than by catastrophic collisions. The main loss mechanisms are inverse Compton scattering on the interstellar radiation field and synchrotron radiation in the Galactic magnetic field. At low energies, inverse Compton losses are described by the Thomson cross section, leading to strong, continuous energy losses. At electron energies of order 40–50 GeV, scattering on ultraviolet photons enters the Klein–Nishina regime, reducing the cross section. Therefore, cosmic-ray electrons are dominated down to energies of tens of GeV [14].

In contrast to CRs, neutrinos interact only via the weak force and have interaction cross sections that are many orders of magnitude smaller. The cross section is measured to be $\sim 10^{-38}$ cm²/nucleon at energies around 1.1 and 3.3 GeV, according to [15]. Therefore, neutrinos interact about $\sim 10^{13}$ times less than hadronic CRs.

Cosmic-ray cross sections depend on particle type; if neutrinos were included as a cosmic-ray component, their interaction cross sections would simply differ from those of other particles. However, because neutrino cross sections are many orders of magnitude smaller, it is reasonable to treat them as a separate category.

C. Detection

CRs are constantly detected and regularly produce unwanted signals, becoming an unavoidable source of experimental noise (e.g. [16]). Consequently, CRs affected experiments long before 1900, although scientists did not correctly interpret these effects until the early 20th century. This could not occur for neutrinos, since their detection is extremely difficult.

Ground-based detectors observe the lateral density distribution of secondary CRs, allowing the shower to be reconstructed and the properties of the primary CR to be studied. In practice, these observatories deploy large arrays of plastic scintillators (which record the passage of charged particles through the blue scintillation light they produce), water-Cherenkov tanks (which measure the Cherenkov light generated by relativistic charged particles in water), fluorescence telescopes (which observe nitrogen fluorescence along the shower track), and, more recently, radio antennas.

Space-based cosmic-ray detectors directly sample primary particles, allowing more accurate measurements at energies up to roughly 100 TeV, above which the rapidly falling flux and payload mass constraints become limiting. These instruments employ magnetic fields to distinguish particles from antiparticles, tracking systems and calorimeters.

Neutrino detectors observe neutrinos indirectly through the secondary charged particles produced in rare weak interactions; these secondaries deposit energy or emit radiation that is measured by the detector. This requires very large target masses, extremely low backgrounds, and complex reconstruction techniques, resulting in much smaller data samples.

D. Origin

Most CRs are believed to originate inside our Galaxy, since particles from other galaxies would not provide the required energy. Because CRs are charged, they undergo magnetic diffusion, which produces nearly isotropic arrival directions and makes identifying their sources one of the main goals of cosmic-ray physics. The dominant theory assumes that charged particles are accelerated in astrophysical shocks and undergo spallation and magnetic scattering before reaching Earth.

Possible origins of CRs include supernova remnants (SNRs), where diffusive shock acceleration (DSA) can explain the observed spectrum up to the knee ($\sim 10^{15}$ eV). Stellar winds and superbubbles may also contribute, potentially up to or beyond the knee and explaining composition anomalies. The Galactic Centre is another possible source, since the inner Galaxy exhibits a harder cosmic-ray spectrum and may act as a PeV accelerator. In addition, gamma-ray observations suggest that the Sun itself may accelerate particles up to TeV energies, called Solar cosmic rays. For UHECRs, likely sources include AGN, GRBs, starburst galaxies, cluster shocks, or fast-spinning neutron stars, with propagation distances limited to ≈ 100 Mpc for protons and even shorter for nuclei.

Cosmic neutrinos originate from the decay of pions and kaons produced where protons and nuclei are accelerated,

so they trace cosmic-ray accelerators. Charged cosmic rays, typically protons, collide with photons or nucleons near their sources, producing pions that decay into neutrinos. Cosmogenic neutrinos arise when UHECRs interact with the CMB or the EBL. Their flux depends on cosmic-ray composition, the dominant interaction channels, and the cosmic evolution of the sources. Current data are still insufficient to determine their origin.

Nevertheless, neutrinos remain key cosmic messengers, as they can propagate without significant absorption or magnetic deflection and trace the origin of CRs.

E. Anisotropy

Cosmic-ray intensities from different directions are not equal; this is known as anisotropy. The amplitude of the anisotropy is small ($\sim 10^{-4}$ – 10^{-3}), but it is statistically significant. The preferred arrival direction of CRs is energy-dependent: around $10^{14.5}$ – 10^{15} eV, the preferred direction reverses.

Simple models in which cosmic rays spread evenly through smooth magnetic fields fail to explain the observed patterns. The data, instead, suggest a more complex scenario, possibly involving local, stochastic sources or structured magnetic fields. At the highest energies (EeV and above), the low flux complicates anisotropy measurements, but current strategies focus on searching for temporal and spatial clustering, tracing the signals back to extragalactic objects. The features of cosmic-ray anisotropy and their energy dependence provide insights about their origin [17].

However, isotropy in cosmic neutrinos is not proven. The statistics of cosmic neutrinos are currently still limited; even after many years, the total number of events is relatively low compared to CRs, so there is a large statistical uncertainty. However, studying neutrino anisotropy could greatly help in using them as cosmic messengers.

F. Electric charge

Recent studies (e.g [18]) argue that, in addition to charged components, they can also include a neutral component of high-energy neutrons with time-dilated lifetimes that propagate ballistically before decaying and act as temporary neutral primaries.

Many efforts, especially from the Pierre Auger Observatory [19], were made to detect UHE neutrons. They searched for small directional clusters of air showers aligned with known Galactic sources. However,

none have been successfully detected yet, but the reason could be their difficulty to be detected due to their expected smaller flux than charged particles and the similarities between neutron-induced air showers and proton-induced ones.

Therefore, one could argue that the definition of CRs should also include uncharged particles, since UHE neutrons, if detected, would be classified as CRs.

V. CONCLUSION

Cosmic neutrinos share several fundamental similarities with CRs, such as their cosmic origin, high energies and the fact that they are fermions and follow approximate power-law spectra. However, they also differ in important ways and hence treating them as separate categories helps avoid confusion, particularly in their cross section, detection methods and in how their energy spectra and origins relate to those of cosmic rays. Moreover, in light of the discussion, their neutral charge alone would not disqualify them from being classified as CRs.

Cosmic rays are not simply “rays” from space. The closest concept to this idea is cosmic radiation, which also includes electromagnetic radiation and neutrinos, even though neutrinos are not “rays”. Cosmic rays instead refer to a specific class of particles.

The first feature was defined historically by their ability to strongly ionize the atmosphere and by their extraterrestrial origin. This criterion already excludes neutrinos, whose interaction cross section is far too small. A second defining feature is that CRs are particles of matter, which excludes electromagnetic radiation such as gamma rays. Also, contrary to usual definitions, they cannot be defined by electric charge, because neutrons can also be CRs. Taken together, these criteria lead to a consistent definition: cosmic rays are matter particles of extraterrestrial origin that interact sufficiently strongly with matter, much more so than neutrinos, to produce detectable effects in the atmosphere.

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